RE501 Assignment 1: Probability Theory – I

Total marks: 50

Instructions

- **Time:** Please submit before the exam. Submissions received after the exam may not be evaluated.
- **Coding:** Some parts need a short program. You may use C, C++, Python, Fortran, MATLAB/Octave, Excel or any other language you are comfortable with. Excel holds about a million rows, fewer than Problem 2 needs.
- **Submission:** Put everything in one folder: your programs, their results (plots, or screenshots of the output) and a scan of your handwritten answers. Compress it as a `.zip` or `.7z` file named after you and the course; for example, Rohit G. Sharma would send `rohit_sharma_re501.zip`. Email it as an attachment to hbnire501@gmail.com.
- **Doing the work yourself:** Please solve the problems on your own, without AI tools. Looking up a concept, or searching the internet for help with your code, is fine. Solutions to such problems may be available online, but copying them without understanding defeats the purpose of this course, because the reasoning you build here is what the rest of the course rests on. An honest attempt, even an incomplete one, is worth far more than a copied answer, and we can discuss it in a session.
- **Discussing with other participants:** You are welcome to discuss the problems with other course participants, but please write your own answers and programs, and specify the name of those you discussed with. This makes it easy for us to evaluate submissions with similar ideas.
- **Copying:** If submissions from two different people are found to be copied, neither will be evaluated.
- **Order of the problems:** Please solve the problems in the given order. Each problem builds on the ideas, and sometimes the results, of the ones before it.
- **Read-through parts ($*$):** Boxes and parts marked $*$ only explain ideas and need no written answer; every other part is an exercise. Read the $*$ parts carefully, as the exercises rest on them. Their marks, shown as [1*], are judged from how your answers use them.
- **Marks:** The marks show the weight of each part. Clear reasoning earns marks even when an answer is incomplete, and the final marks may be scaled across all submissions.

Problem 1: Making exponential random numbers **[4 marks]**

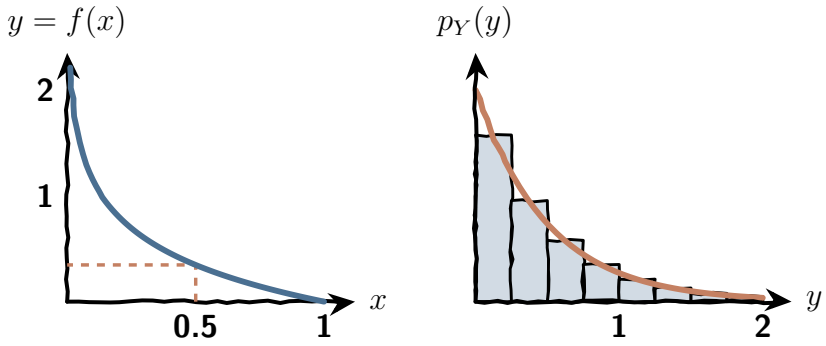
Let $X \sim U(0,1)$ and

$$Y = f(X) = -\frac{1}{\lambda} \ln X.$$

Then $Y \sim \mathrm{Exp}(\lambda)$.

Write a program that generates 100 000 values of X and converts each of them to $Y = f(X)$, for $\lambda = 2$. *Congratulations: you have just learned how to sample an exponential distribution, the first step of a Monte Carlo simulation of particle transport.*

Check that the result really is exponential. Divide the range $0 \le y < 2$ into 50 bins of width $\Delta y = 0.04$ and count the values of Y in each bin. Plot count$/(100\,000\,\Delta y)$ against y and compare it with $p_Y(y) = \lambda e^{-\lambda y}$.



Left: the function f turns each uniform number x into y. Right: for $\lambda = 2$ the histogram of y follows $\lambda e^{-\lambda y}$.

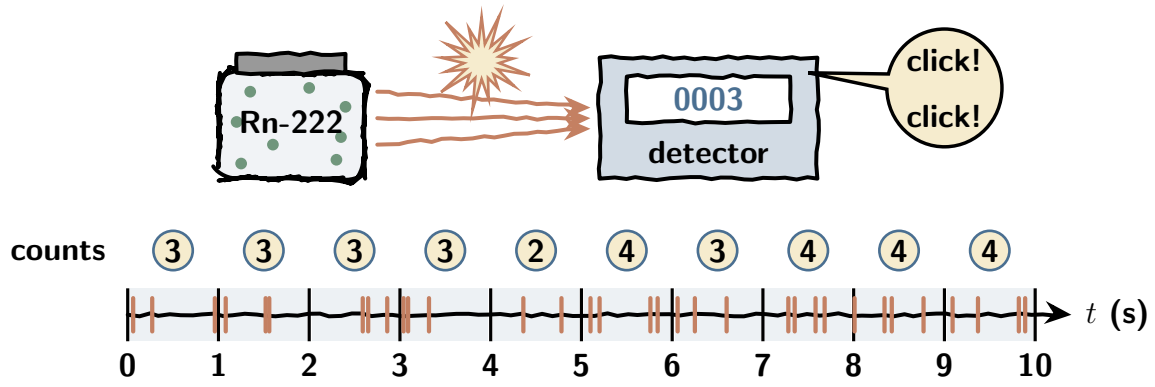
Problem 2: Counting radioactive decays **[8 marks]**

A sealed vial holds $N = 2 \times 10^6$ atoms of Rn-222 (half-life 3.82 days) at $t = 0$. It sits in front of a detector that counts every Rn-222 decay. Ignore the decays of the daughter nuclei.

(a) Find the decay constant λ and the probability of a nuclide decaying in a second. Estimate the average number of Rn-222 decays in the vial in the first second. After 1 second, what is the change in number of Rn-222 nuclides in the vial? So what should be the average number of decays in the next second? **[2]**

(b) Each atom present at $t = 0$ decays at a random time $T \sim \text{Exp}(\lambda)$. Use your program from Problem 1 to sample a decay time for each of the N atoms. **[2]**

(c) Keep the atoms that decay within the first 60 s and count them in each 1 s bin. Find the mean, the variance and the variance-to-mean ratio of the 60 counts. What should this ratio be, and why? **[4]**

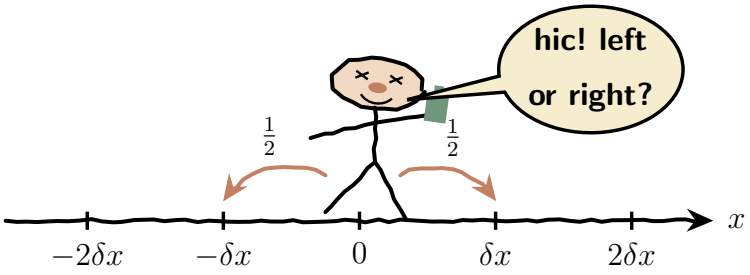


The first 10 s of one simulated minute. Each orange mark is a decay; the circled numbers are the counts in each 1 s bin.

Problem 3: The drunkard's walk **[22 marks]**

A drunkard starts at the origin and keeps taking steps.

(a) In one dimension, every step has the same length δx, to the left or right with equal probability. Let X be his position. For (i) one step and (ii) two steps, make the probability distribution table of X (each value and its probability) and find $\mathrm{E}(X)$ and $\mathrm{E}(X^2)$. Explain why $\mathrm{E}(X) = 0$ after any number of steps. **[2]**



In general, neither the step nor the waiting time is fixed. His position at time t then depends on how many steps he has taken by t, the size and sign of each step, and the waiting time before each step. We assume the steps are independent, with values drawn from some distribution. **You will now show that if the waiting time is memoryless and the step size follows a symmetric distribution with zero mean and finite variance, the probability of finding the drunkard at (x, t) follows the diffusion equation.**

∗ **General case setup.** (*read through*) **[1*]**

Assumptions.

- The waiting time W before each step is exponential with a constant mean τ: $W \sim \text{Exp}(1/\tau)$.
- Each step S has a PDF $p_S(s)$ with $\text{E}(S) = 0$ and finite $\text{E}(S^2)$.

Notation.

W_k, τ	waiting time before the k-th step; average waiting time $\tau = \text{E}(W)$
S, S_k, $p_S(s)$	step along one axis; the k-th such step; its PDF
S_x, S_y, S_z	components of one step in three dimensions
L, L_{rms}	length of one step in any direction, $L^2 = S_x^2+S_y^2+S_z^2$; $L_{rms} = \sqrt{\text{E}(L^2)}$
n, $\bar{S}_n$	number of steps by time t, $n \approx t/\tau$; sample mean of the n steps
X_t, $p_X(x,t)$	displacement along one axis from the start at time t; its PDF
R_t, R_{rms}	displacement from the start in three dimensions at time t, $R_t^2 = X_t^2 + Y_t^2 + Z_t^2$; $R_{rms} = \sqrt{\text{E}(R_t^2)}$
$p(x,y,z,t)$	PDF of the position in three dimensions
D	diffusion coefficient

Let the sample space Ω be the set of all possible walks

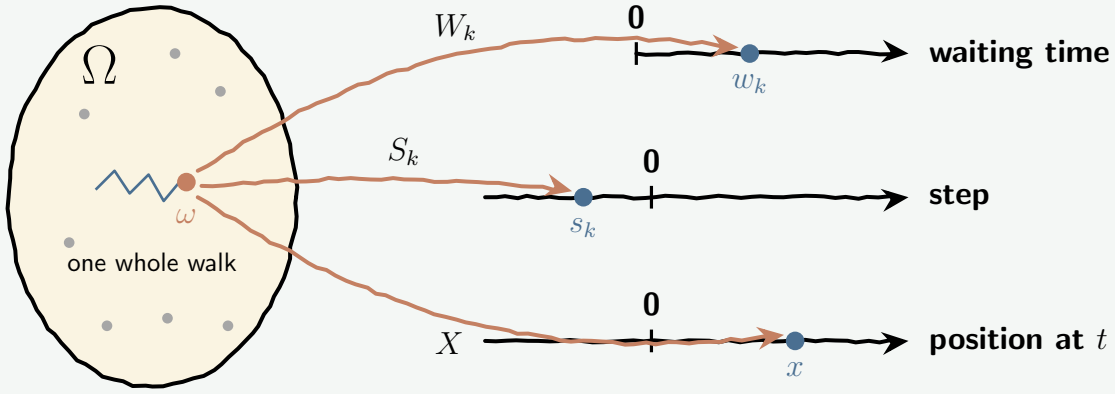
$$\omega = \big((w_1, s_1), (w_2, s_2), \dots\big),$$

where w_k is the waiting time before the k-th step and s_k is the k-th step. On Ω define the random variables

- the k-th waiting time, $W_k(\omega) = w_k$, with $W_k \sim \text{Exp}(1/\tau)$; after n steps, $k \in \mathbb{I}[1, n]$, the integers from 1 to n;

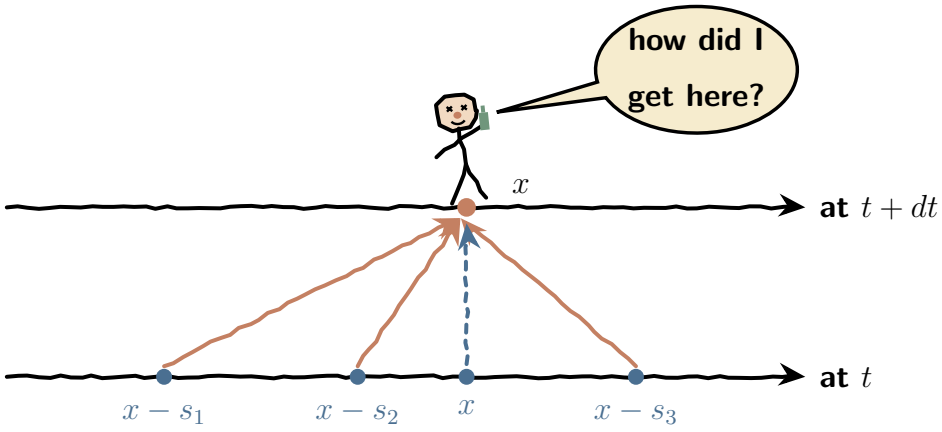
- the k-th step, $S_k(\omega) = s_k$, with PDF $p_S(s)$;

- the position at time t, $X_t(\omega) = x$, the sum of the steps taken by t (those with $w_1 + \cdots + w_k \le t$). Its PDF is $p_X(x,t)$, with t a parameter, not a random variable: $\Pr_X(x, x+dx; t) = p_X(x,t)\,dx$.



Each outcome ω is one whole walk. The three random variables read off three different numbers from it.

(b) To be at x at time $t+dt$, either he was at x at time t and did not step during dt, or he was in a small interval around $x-s$ and took a step between s and $s+ds$. From these probabilities, write $p_X(x, t+dt)$ in terms of $p_X(x,t)$, $p_X(x-s,t)$ and $p_S(s)$, dropping terms above first order in ds. State which events are mutually exclusive and which are independent. **[5]**



Orange: he takes a step s during dt. Blue dashed: he does not step.

(c) Let the steps s be small and $dt \to 0$. Expand $p_X(x-s,t)$ as a Taylor series in s, up to the first order that survives integration over s, and show that

$$\frac{\partial p_X}{\partial t} = D\,\frac{\partial^2 p_X}{\partial x^2},$$

where

$$D = \frac{\mathrm{E}(S^2)}{2\tau}.$$

In three dimensions, a step has independent components S_x, S_y, S_z, each distributed like S, and length L, with $L^2 = S_x^2 + S_y^2 + S_z^2$. Show that $\mathrm{E}(L^2) = 3\ \mathrm{E}(S^2)$, so $D = \mathrm{E}(S^2)/2\tau = \mathrm{E}(L^2)/6\tau$. **[3]**

(d) Assume $p_X(x,t)$ decays faster than $1/x^3$ as $x \to \pm\infty$. Multiply the diffusion equation by x and integrate over all x to show that $\mathrm{E}(X_t) = \mathrm{E}(X_0) = 0$. Do the same with x^2 to show that $\sigma_X^2(t) = \mathrm{E}(X_t^2) = 2Dt$, so $D = \mathrm{E}(X_t^2)/2t$. [*Hint*: integrate by parts.] In three dimensions, the distance R_t from the origin has $R_t^2 = X_t^2 + Y_t^2 + Z_t^2$, with each coordinate like X_t, so $D = \mathrm{E}(R_t^2)/6t$. Check that, for one step ($t \to \tau$, $X_t \to S$, $R_t \to L$), these agree with (c). D is the same in one and three dimensions; the factor 3 comes from $\mathrm{E}(R_t^2) = 3\ \mathrm{E}(X_t^2)$ and $\mathrm{E}(L^2) = 3\ \mathrm{E}(S^2)$.

$$\boxed{D = \frac{\mathrm{E}(R_t^2)}{6t} = \frac{\mathrm{E}(L^2)}{6\tau}}$$

What would D be, in terms of $\mathrm{E}(R_t^2)$ and $\mathrm{E}(L^2)$, if the movement were confined to two dimensions? **[4]**

(e) Now look at the same result through the steps. In one dimension, after n steps $X_t = S_1 + S_2 + \cdots + S_n = n\,\bar{S}_n$, where $\bar{S}_n$ is the sample mean of the steps. For large n, use the CLT to write the distribution

of $\bar{S}_n$. **[1]**

(f) Taking $n \approx t/\tau$, argue that $X_t = n\,\bar{S}_n$ is approximately $\mathcal{N}(0,\, 2Dt)$, with PDF

$$p_X(x,t) = \frac{1}{\sqrt{4\pi Dt}}\, e^{-x^2/4Dt}.$$

Check that it satisfies the diffusion equation of (c). **[2]**

(g) In three dimensions, the steps along x, y and z are i.i.d. Following (b) and (c), show that the PDF $p(x,y,z,t)$ of his position satisfies

$$\boxed{\frac{\partial p}{\partial t} = D\left(\frac{\partial^2 p}{\partial x^2} + \frac{\partial^2 p}{\partial y^2} + \frac{\partial^2 p}{\partial z^2}\right)}$$

[1]

(h) The three coordinates are independent and, by symmetry, each has the PDF of X_t from (f). Hence argue that

$$\boxed{p(x,y,z,t) = \frac{1}{(4\pi Dt)^{3/2}}\, e^{-(x^2+y^2+z^2)/4Dt}},$$

and check that it satisfies the diffusion equation of (g). **[1]**

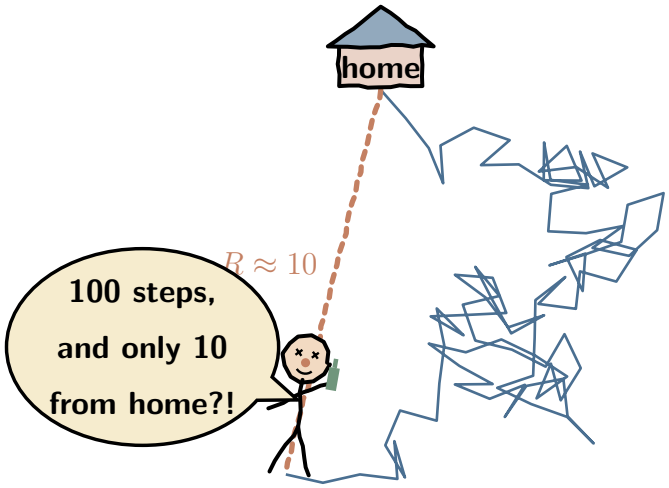
Congratulations: you have just derived the diffusion equation from probability theory. A few points are worth noting.

∗ **Note: what the diffusion equation really needs.** (*read through*) **[1*]**

- The waiting time must be exponential (memoryless), so that the probability of a step during dt is dt/τ.
- The steps may follow any symmetric distribution with $\mathrm{E}(S) = 0$ and

finite $\mathrm{E}(S^2)$, independent of each other and of the waiting times.
Any such distribution gives the same diffusion equation, with $D = \mathrm{E}(L^2)/6\tau$.

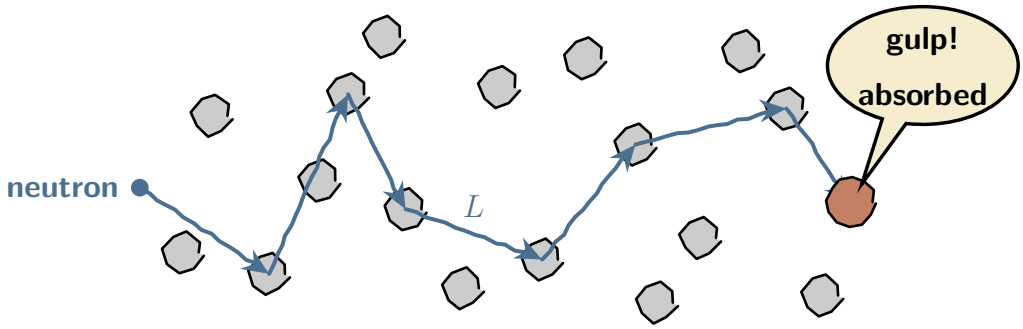
$*$(i) *RMS distance and step size* (*read through*): From (c) and (d), the *rms* step length $L_{rms} = \sqrt{\mathrm{E}(L^2)} = \sqrt{6D\tau}$ is constant, while the *rms* distance from the origin after n steps ($t = n\tau$) is $R_{rms} = \sqrt{\mathrm{E}(R_t^2)} = \sqrt{6Dt} = L_{rms}\sqrt{n}$. So, although each step is $\sim L_{rms}$, he ends up only $\sim L_{rms}\sqrt{n}$, not $\sim nL_{rms}$, from the origin: for $L_{rms} = 1$ and $n = 100$, he walks about 100 but ends only about 10 from home. **[1*]**



One walk of 100 steps, each of length 1.

Problem 4: From the drunkard to the neutron [16 marks]

A neutron in a large block of moderator moves like the drunkard of Problem 3 in three dimensions. It flies in a straight line, hits a nucleus, and flies off in a new direction. Now and then a nucleus swallows it instead, and its walk ends.



Each flight between collisions is one step of the walk. The orange nucleus absorbs the neutron.

∗ **Assumptions.** (*read through*) **[1*]**

- There is no source of neutrons.
- Only scattering and absorption of neutrons are possible in the medium.
- All neutrons have the same speed v throughout their travel.
- After each collision the neutron flies off in any direction, all directions being equally probable.
- At a collision the neutron can be absorbed, with a very low probability Σ_a/Σ_t. Since there is only scattering and absorption possible, this leads to $\Sigma_s \gg \Sigma_a$ and $\Sigma_t \approx \Sigma_s$.
- The distance L it travels before its next collision is exponentially distributed, $L \sim \text{Exp}(\Sigma_t)$, with mean $1/\Sigma_t \approx 1/\Sigma_s = \lambda_s$, the mean free path.

∗ **Discussion.** (*read through*) **[1*]**

- *Constant energy.* We assume a neutron keeps its energy, and so its speed v, after every scattering. This does not happen in practice: the energy of a neutron changes during collision. It is a simplification that keeps the model easy.
- *Exponential step size.* The probability that a neutron collides within the next short distance δx is $\Sigma_t\,\delta x$, however far it has already flown. This is a memoryless process, so the flight length, the distance it travels before its next collision, is exponential: $L \sim \mathrm{Exp}(\Sigma_t)$. There is no longest flight: most flights are shorter than $\lambda_t = 1/\Sigma_t$, but now and then one is much longer.

  Because the speed is constant, the time between collisions, $W = L/v$, is also exponential, $W \sim \mathrm{Exp}(1/\tau)$, just like the drunkard's waiting time, with

  $$\lambda_t = v\,\tau.$$

  So for the neutron the exponential waiting time is a consequence of the medium, not an assumption of its own.
- *Absorption.* The drunkard never stopped walking. The neutron, at a collision, can be absorbed instead of flying on, and its walk ends there. The probability of absorption in a time interval δt is $\Sigma_a v\,\delta t$.

(a) Show that the random-walk coefficient of the neutron is $\mathcal{D} = v\lambda_s/3$. *Beware:* $\mathcal{D}$ is not the neutron diffusion coefficient. The neutron diffusion coefficient is defined in terms of the neutron flux, not the neutron concentration. **[2]**

(b) Let us take $N = N(0)$ neutrons in total at $t = 0$, each undergoing diffusion. We can define $n(\mathbf{r}, t) = N\, p(\mathbf{r}, t)$, where $\mathbf{r} = (x, y, z)$ and p is the probability density of the particle (its unit is per cc). Then $n(\mathbf{r}, t)$ is the probable number of neutrons per unit volume, and $n(\mathbf{r}, t)\, d\mathbf{r}$ is the expected number of neutrons in the small volume $d\mathbf{r}$ around $\mathbf{r}$ at time t. Let $p_L(\boldsymbol{\ell})$ be the PDF of one flight $\boldsymbol{\ell}$ (its length and direction together). Repeat the argument of Problem 3(b) for n and write $n(\mathbf{r}, t + dt)$ in terms of $n(\mathbf{r}, t)$, $n(\mathbf{r} - \boldsymbol{\ell}, t)$ and $p_L(\boldsymbol{\ell})$.

Hint: as with the drunkard's steps, treat each flight $\boldsymbol{\ell}$ as a jump made at a collision. A neutron collides during dt with probability dt/τ: it is absorbed with probability $\Sigma_a v\, dt$ and scattered with probability $dt/\tau - \Sigma_a v\, dt$. A neutron found at $\mathbf{r}$ at time $t + dt$ either was at $\mathbf{r}$ and did not collide, or was at $\mathbf{r} - \boldsymbol{\ell}$, was scattered, and flew $\boldsymbol{\ell}$. A neutron that is absorbed is simply lost.

Which factor is new compared with Problem 3(b)?

Then, just like in the drunkard's walk problem, show that

$$\frac{\partial n(\mathbf{r}, t)}{\partial t} = \mathcal{D}\, \nabla^2 n(\mathbf{r}, t) - \Sigma_a v\, n(\mathbf{r}, t), \qquad \nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}.$$

Integrate over all $\mathbf{r}$ and show that the total number of neutrons changes with time as

$$N(t) = N(0)\, e^{-\Sigma_a v t}.$$

[6]

∗ **Note: where does the probability go?** (*read through*)
At time t a neutron is either at some point $\mathbf{r}$ or already absorbed, so $\int p(\mathbf{r},t)\,d\mathbf{r} + \Pr(\text{absorbed by } t) = 1$ at every t: total probability is conserved. Here p covers only the neutrons still flying, so $\int p\,d\mathbf{r}$ decreases as probability flows into "absorbed". For a neutron started at the origin, $p(\mathbf{r},t) = e^{-\Sigma_a v t}\,(4\pi\mathcal{D}t)^{-3/2}\,e^{-|\mathbf{r}|^2/4\mathcal{D}t}$. Given that the neutron is still flying, its position is the normalised Gaussian. Absorption only scales it down and does not change its shape.

(c) The neutron flux is defined as $\phi(\mathbf{r},t) = n(\mathbf{r},t)\,v = N\,p(\mathbf{r},t)\,v$, with unit per cm^2 per s, and the diffusion coefficient is $D = \mathcal{D}/v$. Show that

$$\boxed{\frac{1}{v}\frac{\partial\phi}{\partial t} - D\nabla^2\phi + \Sigma_a\phi = 0}$$

This is the one-group neutron diffusion equation without a source, which you have built from probability theory. Show that $D = \lambda_s/3$. Where does the factor 3 come from? [*Hint*: what would D be if the neutrons moved in only one or two dimensions?] What are the units of $\mathcal{D}$ and of D? **[2]**

(d) Consider a neutron started at the origin at $t = 0$. Since the probability that it will not get absorbed in finite time t is $e^{-\Sigma_a v t}$, find the average time it flies before getting absorbed. Find the root mean square distance it travels from its starting point before getting absorbed. [*Hint*: its mean squared distance from the origin grows as $6\mathcal{D}t$.] Show that

$$R_{rms} = \sqrt{6}\,L_d, \quad \text{where} \quad L_d^2 = \frac{D}{\Sigma_a}.$$

L_d is called the *diffusion length*. **[4]**